# HOUEHA Létondé Précieux Fanès

houeha@aims.ac.za | (+229) 01 94 745 373 | Cape Town, South Africa | Fanès Houeha

## Profile

Statistician-Economist Engineer specialized in Applied Data Science, currently pursuing an AI for Science Master's on a full Google DeepMind scholarship. Motivated to apply data, AI, and digital tools to advance evidence-based, inclusive development outcomes.

## Education

**African Institute for Mathematical Sciences (AIMS South Africa)** Muizenberg, South Africa
*M.Sc. in AI for Science – Google DeepMind Scholar (Full Scholarship)* *Sep. 2026 – Present*

- Selected for a fully-funded, research-oriented Master's programme applying artificial intelligence methods to scientific and development challenges.

**National School of Applied Economics and Management – University of Abomey-Calavi** Cotonou, Benin
*Statistician-Economist Engineer (ISE) – Master's Level* *Oct. 2023 – Jul. 2026*

- Graduated with **Honors**
- Thesis: Machine Learning Prediction of the Vessel Turnaround Time at the Autonomous Port of Cotonou **(Highest Honors)**.

**National School of Applied Economics and Management – University of Abomey-Calavi** Cotonou, Benin
*Bachelor in Project Planning and Management* *Oct. 2020 – Sep. 2023*

- Graduated with **Honors** (Major: Planning & Management).
- Thesis: Digital Transformation of Monitoring & Evaluation Systems: Focus on the Sustainable Livelihoods Project **(Highest Honors)**.

## Professional Experience

**Central Bank of West African States (BCEAO)** | **Cotonou Agency** Cotonou, Benin
*Research Assistant Intern* *Jun. – Aug. 2026*

- Selected for the 2026 edition of BCEAO's internship programme.
- Evaluated the impact of the deployment of the interoperable instant payment platform on the digital payments ecosystem in Benin, informing the regional central bank's payment systems modernization agenda.

**Cotonou Port Authority (PAC)** | **Financial Audit & Internal Control Department** Cotonou, Benin
*Data Scientist Intern* *Mar. – Jun. 2026*

- Completed an academic internship at the PMO and Reporting Unit of the Department.
- Implemented two different ML models using Random Forest, LightGBM, CatBoost and XGBoost algorithms to predict anchorage time quay operations duration, the two most variables components of the Vessel Turnaround Time at the Port.

**Ministry of Development and Government Action Coordination** Cotonou, Benin
*Data Analyst & Policy Evaluation Intern* *May – Aug. 2023*

- Conducted quantitative analysis of semi-annual budgetary performance indicators for national public policy impact evaluation, directly informing ministerial decision-making.
- Managed full research cycle for a digitalization influence study – from field data collection (ODK) to statistical analysis (Stata) and policy recommendations – within a GEF/UNDP project (Sustainable Livelihoods Project).

## Selected Research Project

**Impact of Electrification on Non-Agricultural Entrepreneurship – Causal Inference (R)** *2026*

- Evaluated the causal effect of electricity grid access on household business creation in Benin using World Bank EHCVM 2021–22 survey data (8,000+ households); addressed selection bias via Propensity Score Matching (PSM) and Logit models – identifying a 15.3% increase in non-agricultural enterprise probability.
- Conducted gender-based heterogeneity analysis, revealing significant disparities in outcomes between male- and female-led businesses, with implications for targeting of social policies.
- Wrote fully documented, reproducible R code with clean data pipelines, indicator construction scripts, and regression output tables.

**AI & Inclusive Growth in Benin: Transmission Channels (R & Python)** *2025*

- Analyzed the impact of digital adoption on inclusive growth using an ARDL cointegration approach on Beninese data (2000–2023).
- Developed a Composite Digital Adoption Index (CIDA) using Principal Component Analysis (PCA) and constructed a Synthetic Inclusive Growth Indicator (ICI) integrating health, education, and income dimensions.
- Identified that while digital impact is currently latent, physical infrastructure remains the primary driver of inclusion, providing evidence-based recommendations for Benin's national AI strategy.

## Leadership & Community Engagement

**U-report Benin (UNICEF Initiative)** Jul. 2025 – Present
*Active Member / Youth Advocate* *Abomey-Calavi, Benin*

- Actively contribute to global consultations on social development, synthesizing youth perspectives to inform high-level policy dialogues and UNICEF strategic initiatives.

**Statistician-Economists Student Association** Feb. 2025 – July 2026
*Communication Unit Coordinator* *Cotonou, Benin*

- Conducted with a team of three the implementation of the first professional website of the association
- Direct with a team of five persons the association's digital transformation strategy, managing the web presence and cross-channel communications (Facebook, LinkedIn) to ensure consistent institutional messaging.
- Coordinate information flows between the executive board and the student body, streamlining internal documentation and external outreach.

**International Conference on Project Management (ICPM)** Mar. 2025 – Present
*Organizing Committee Member* *Cotonou, Benin*

- Support executive coordination and logistical planning for a major international forum, ensuring seamless engagement with global development experts and stakeholders.

## Technical Skills

**Machine Learning & AI**: Ensemble Methods (Random Forest, XGBoost, LightGBM, CatBoost), Bayesian Optimization (Optuna), Quantile Regression, Unsupervised Learning, Scikit-Learn, Transformers.
**Causal Inference & Econometrics**: Causal ML (Causal Forests), Double/Debiased Machine Learning, Impact Evaluation (DiD, Matching, RCTs), Time Series (ARDL, VAR), Panel Data Analysis.
**Programming & Tools**: Python (Advanced), R (Advanced), SQL (Relational Databases), Stata, Git/GitHub, Streamlit, LaTeX.
**Data Engineering & Deployment**: Model Deployment, Pipeline Optimization, Field Data Architectures (ODK, KoboToolbox).

## Certifications & Professional Training

**Train a Small Language Model** Sept. 2026
*Google Deepmind*

**Write and Publish a Scientific Article** Feb. 2026
*Research for Development Institute (IRD) & France Université Numérique (FUN)*

**Qualitative Data Management and Analysis for Monitoring and Evaluation (M&E)** Jan. 2026
*IDEAL, United States Agency for International Development (USAID)*

**AI for Social Impact** Dec. 2025
*Asia Development Bank Institute*

**McKinsey Forward Program** Dec. 2024
*McKinsey & Company*

**Certificate of Competence in Oral Communication** Oct. 2023
*Master Class MC International*

## Awards & Distinctions

**2nd Prize, "Œconomicarum" University Competition** May 2022
*Inter-departmental economics public speaking and general knowledge contest*

## Academic Conferences & Seminars

**Deep Learning IndabaX Benin** Dec. 2025
*Participant | Benin Excellence Library*

- Sessions on Active Learning and Federated Learning for privacy-preserving ML applications.

**6th Scientific Days of the Beninese Economy** Nov. 2025
*Participant | Ministry of Economy and Finance*

- **Theme:** Institutions and Prosperity of Nations – with special participation of Nobel Laureate in Economics, Prof. James A. Robinson.

**ACP Summer School 2025 – Constraint Programming for Sustainable Development** Aug. 2025
*Participant | University of Abomey-Calavi*

- Co-facilitated by Prof. Pierre Schaus, Dr. Hélène Verhaeghe, and Dr. John Aoga (UCLouvain).

## Languages

**French**: Native Speaker
**English**: Professional Working Proficiency – **TOEFL iBT: 91/120 (Valid since Nov. 4, 2025)**